

Netflix Movies and TV Shows Data Analysis

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Introduction

This tiny project analyses Netflix catalogue data to understand the distribution of Movies and TV Shows, release-year patterns, ratings, countries, and genres. It demonstrates an end-to-end R workflow: data import, cleaning, statistical exploration, plots, interpretation, and a reproducible conclusion.

Project Definition

The Netflix Movies and TV Shows Data Analysis project is a data analysis project developed using the R programming language. The project explores a Netflix catalogue dataset to understand the distribution of Movies and TV Shows and examine patterns related to release years, ratings, countries, and genres.

The project includes data importing, preprocessing, statistical analysis, data visualization, interpretation of results, and a reproducible conclusion.

Objectives

The main objectives of this project are:

1. To analyze the Netflix Movies and TV Shows dataset using R.
2. To identify the distribution of Movies and TV Shows.
3. To examine release-year patterns.
4. To analyze the content ratings.
5. To identify frequently represented production countries.
6. To identify frequently occurring genres.
7. To visualize the results using graphs.
8. To interpret the findings from the analysis.

Dataset Description

The dataset contains information about Netflix Movies and TV Shows. The main attributes used in this analysis are:

Attribute	Description
<code>show_id</code>	Unique identifier of the title
<code>type</code>	Indicates whether the title is a Movie or TV Show

Attribute	Description
title	Name of the title
director	Director of the title
cast	Cast information
country	Production country or countries
date_added	Date the title was added
release_year	Original release year
rating	Content rating
duration	Movie duration or number of seasons
listed_in	Genres or categories
description	Description of the title

Tools and Technologies

- R
- RStudio
- R Markdown
- dplyr
- ggplot2
- stringr
- tidyr
- knitr

Data importing

Download a Netflix titles CSV file and save it as `netflix_titles.csv` in the same folder as this `.Rmd` file. This report expects the commonly used columns: `show_id`, `type`, `title`, `director`, `cast`, `country`, `date_added`, `release_year`, `rating`, `duration`, `listed_in`, and `description`.

```
netflix <- read.csv("netflix_titles.csv", stringsAsFactors = FALSE, na.strings = c("", "NA"))
head(netflix)
```

```
##  show_id  type      title      director      cast
## 1      s1  Movie The Silent City Asha Mehta Riya Shah
## 2      s2 TV Show Code Detectives Rohan Patel Jay Kumar
## 3      s3  Movie   Ocean Light   Maya Rao Anika Bose
## 4      s4  Movie Midnight Signal Arjun Shah Vikram Das
## 5      s5 TV Show  Family Kitchen Nina Thomas Leena Roy
## 6      s6  Movie   Future Earth   Sam Lee  Kai Chen
##
##              country      date_added release_year rating  duration
## 1              India September 9, 2019          2018  TV-14    98 min
## 2      India, United States January 12, 2020          2019  TV-PG 2 Seasons
## 3              United States March 15, 2021          2020  PG-13   112 min
## 4              India      July 1, 2021          2021  TV-MA   105 min
## 5              United Kingdom August 20, 2022          2022  TV-G    1 Season
## 6 South Korea, United States November 11, 2022          2022  TV-14   120 min
##
##              listed_in      description
## 1      Dramas, International Movies      A fictional drama.
## 2      TV Dramas, International TV Shows A technology mystery series.
```

```
## 3 Documentaries, International Movies A documentary about oceans.
## 4 Thrillers, International Movies A suspense thriller.
## 5 Reality TV, International TV Shows A cooking competition.
## 6 Sci-Fi & Fantasy, Dramas A science fiction story.
```

```
str(netflix)
```

```
## 'data.frame': 10 obs. of 12 variables:
## $ show_id : chr "s1" "s2" "s3" "s4" ...
## $ type : chr "Movie" "TV Show" "Movie" "Movie" ...
## $ title : chr "The Silent City" "Code Detectives" "Ocean Light" "Midnight Signal" ...
## $ director : chr "Asha Mehta" "Rohan Patel" "Maya Rao" "Arjun Shah" ...
## $ cast : chr "Riya Shah" "Jay Kumar" "Anika Bose" "Vikram Das" ...
## $ country : chr "India" "India, United States" "United States" "India" ...
## $ date_added : chr "September 9, 2019" "January 12, 2020" "March 15, 2021" "July 1, 2021" ...
## $ release_year: int 2018 2019 2020 2021 2022 2022 2023 2021 2024 2024
## $ rating : chr "TV-14" "TV-PG" "PG-13" "TV-MA" ...
## $ duration : chr "98 min" "2 Seasons" "112 min" "105 min" ...
## $ listed_in : chr "Dramas, International Movies" "TV Dramas, International TV Shows" "Documentary" ...
## $ description : chr "A fictional drama." "A technology mystery series." "A documentary about oceans."
```

```
summary(netflix)
```

```
##      show_id      type      title      director      cast
## Length :10 Length :10 Length :10 Length :10 Length :10
## N.unique :10 N.unique : 2 N.unique :10 N.unique :10 N.unique : 9
## N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0 N.blank : 0
## Min.nchar: 2 Min.nchar: 5 Min.nchar:11 Min.nchar: 7 Min.nchar: 7
## Max.nchar: 3 Max.nchar: 7 Max.nchar:17 Max.nchar:11 Max.nchar:10
##
##      country      date_added release_year      rating      duration
## Length :10 Length :10 Min. :2018 Length :10 Length :10
## N.unique : 7 N.unique :10 1st Qu.:2020 N.unique : 6 N.unique : 9
## N.blank : 0 N.blank : 0 Median :2022 N.blank : 0 N.blank : 0
## Min.nchar: 5 Min.nchar:12 Mean :2021 Min.nchar: 4 Min.nchar: 6
## Max.nchar:26 Max.nchar:17 3rd Qu.:2023 Max.nchar: 5 Max.nchar: 9
## Max. :2024
##
##      listed_in      description
## Length :10 Length :10
## N.unique :10 N.unique :10
## N.blank : 0 N.blank : 0
## Min.nchar:21 Min.nchar:15
## Max.nchar:35 Max.nchar:29
##
```

Preprocessing and cleaning

This section standardises text values, converts dates, removes duplicate titles, and removes records with invalid release years. Missing director, cast, country, rating, and duration values are replaced by **Unknown** so they can be counted rather than silently excluded.

```

required_columns <- c("show_id", "type", "title", "director", "cast", "country",
                      "date_added", "release_year", "rating", "duration", "listed_in", "description")
missing_columns <- setdiff(required_columns, names(netflix))
if (length(missing_columns) > 0) {
  stop("The CSV is missing: ", paste(missing_columns, collapse = ", "))
}

netflix_clean <- netflix %>%
  mutate(
    across(where(is.character), ~trimws(.x)),
    date_added = as.Date(date_added, format = "%B %d, %Y"),
    year_added = as.integer(format(date_added, "%Y")),
    release_year = as.integer(release_year),
    type = factor(type),
    rating = ifelse(is.na(rating) | rating == "", "Unknown", rating),
    country = ifelse(is.na(country) | country == "", "Unknown", country),
    listed_in = ifelse(is.na(listed_in) | listed_in == "", "Unknown", listed_in),
    director = ifelse(is.na(director) | director == "", "Unknown", director),
    duration = ifelse(is.na(duration) | duration == "", "Unknown", duration)
  ) %>%
  distinct(show_id, .keep_all = TRUE) %>%
  filter(!is.na(title), title != "", !is.na(release_year), release_year >= 1900, release_year <= 2100)

colSums(is.na(netflix_clean))

```

```

##      show_id      type      title      director      cast      country
##          0          0          0          0          0          0
##  date_added release_year      rating      duration      listed_in      description
##          0          0          0          0          0          0
##   year_added
##          0

```

```

cleaned_preview <- netflix_clean %>%
  select(show_id, type, title, country, release_year, rating)

kable(
  head(cleaned_preview),
  caption = "First six cleaned records"
)

```

Table 1: First six cleaned records

show_id	type	title	country	release_year	rating
s1	Movie	The Silent City	India	2018	TV-14
s2	TV Show	Code Detectives	India, United States	2019	TV-PG
s3	Movie	Ocean Light	United States	2020	PG-13
s4	Movie	Midnight Signal	India	2021	TV-MA
s5	TV Show	Family Kitchen	United Kingdom	2022	TV-G
s6	Movie	Future Earth	South Korea, United States	2022	TV-14

Statistical exploration

```
overview <- netflix_clean %>%  
  summarise(  
    Total_titles = n(),  
    Movies = sum(type == "Movie"),  
    TV_Shows = sum(type == "TV Show"),  
    Earliest_release = min(release_year),  
    Latest_release = max(release_year),  
    Unique_ratings = n_distinct(rating),  
    Unique_countries = n_distinct(country)  
  )  
kable(overview, caption = "Netflix catalogue overview")
```

Table 2: Netflix catalogue overview

Total_titles	Movies	TV_Shows	Earliest_release	Latest_release	Unique_ratings	Unique_countries
10	6	4	2018	2024	6	7

```
type_count <- netflix_clean %>% count(type, sort = TRUE)  
rating_count <- netflix_clean %>% count(rating, sort = TRUE) %>% slice_head(n = 10)  
kable(type_count, caption = "Titles by content type")
```

Table 3: Titles by content type

type	n
Movie	6
TV Show	4

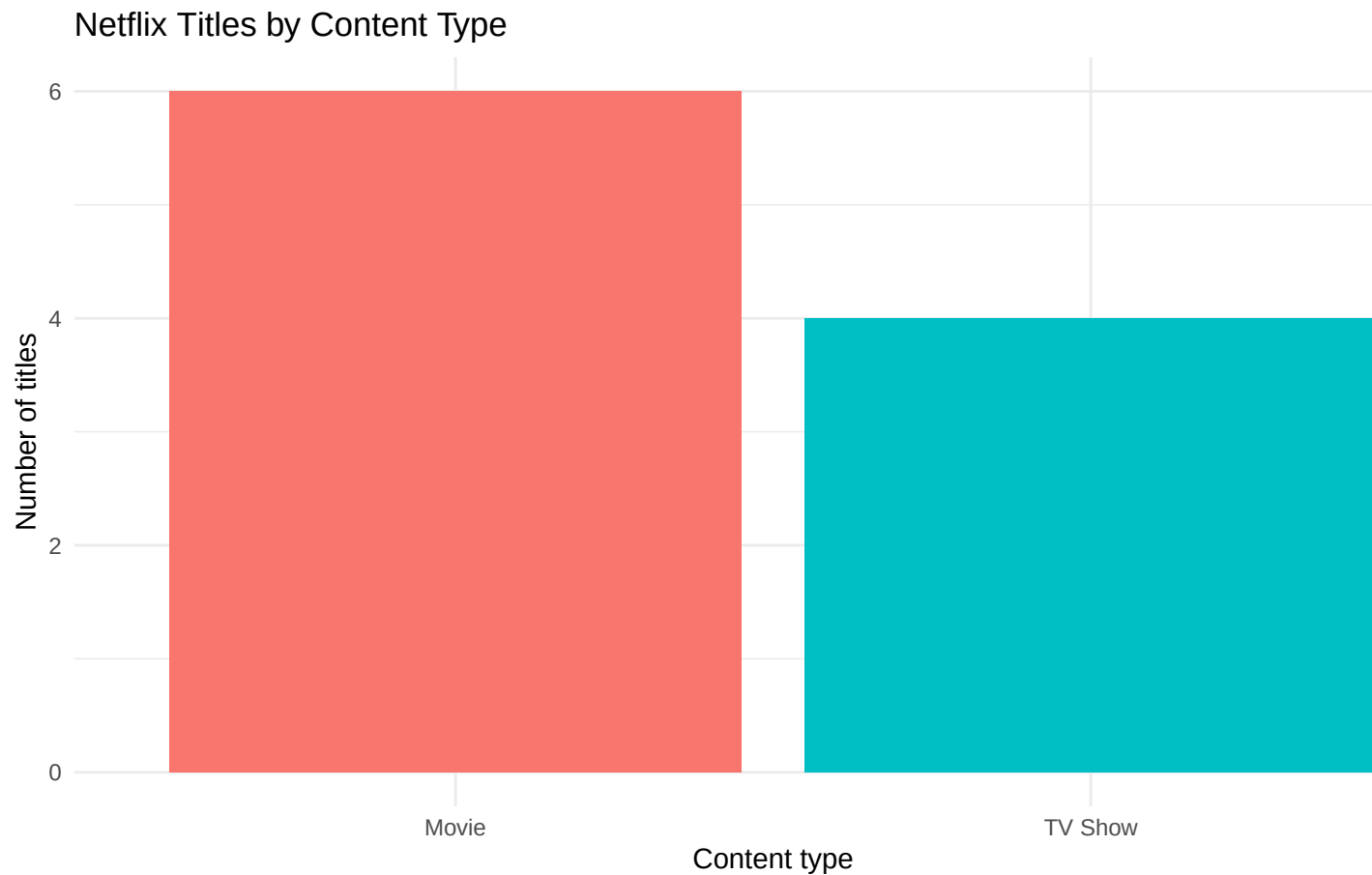
```
kable(rating_count, caption = "Ten most frequent ratings")
```

Table 4: Ten most frequent ratings

rating	n
TV-14	4
TV-PG	2
PG-13	1
TV-G	1
TV-MA	1
TV-Y7	1

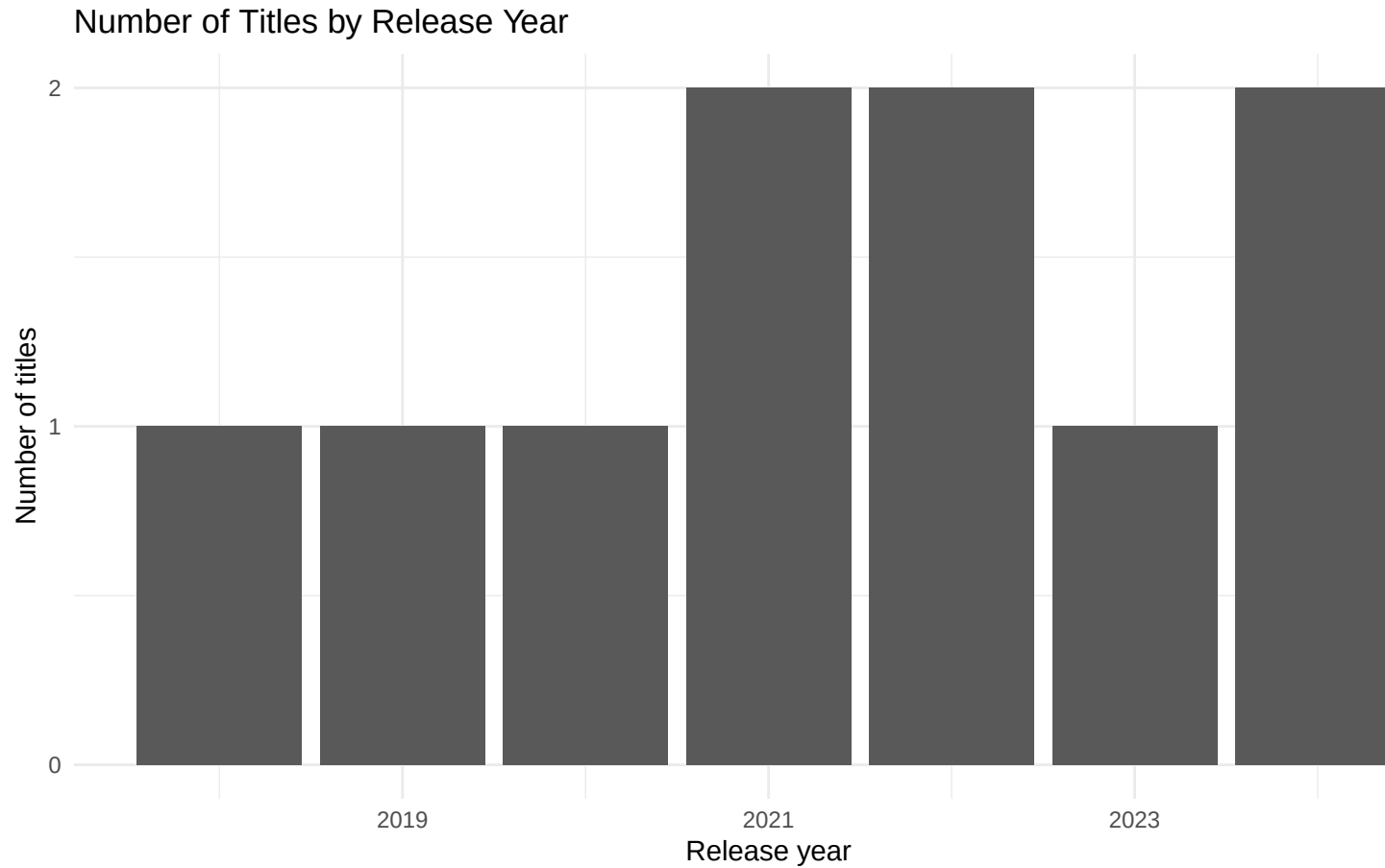
Graphs and plots

```
ggplot(netflix_clean, aes(x = type, fill = type)) +
  geom_bar() +
  labs(title = "Netflix Titles by Content Type", x = "Content type", y = "Number of titles") +
  theme_minimal() + theme(legend.position = "none")
```



```
release_counts <- netflix_clean %>%
  count(release_year)

ggplot(release_counts, aes(x = release_year, y = n)) +
  geom_col() +
  labs(
    title = "Number of Titles by Release Year",
    x = "Release year",
    y = "Number of titles"
  ) +
  scale_y_continuous(
    breaks = seq(0, max(release_counts$n), by = 1)
  ) +
  theme_minimal()
```



The bar chart shows the distribution of titles by original release year. A lower number of titles in some years may reflect the limited size of the dataset rather than an actual production decline.

```
top_countries <- netflix_clean %>%
  filter(country != "Unknown") %>%
  mutate(country = str_split(country, "\\s*")) %>%
  tidyr::unnest(country) %>%
  count(country, sort = TRUE) %>%
  slice_head(n = 10)

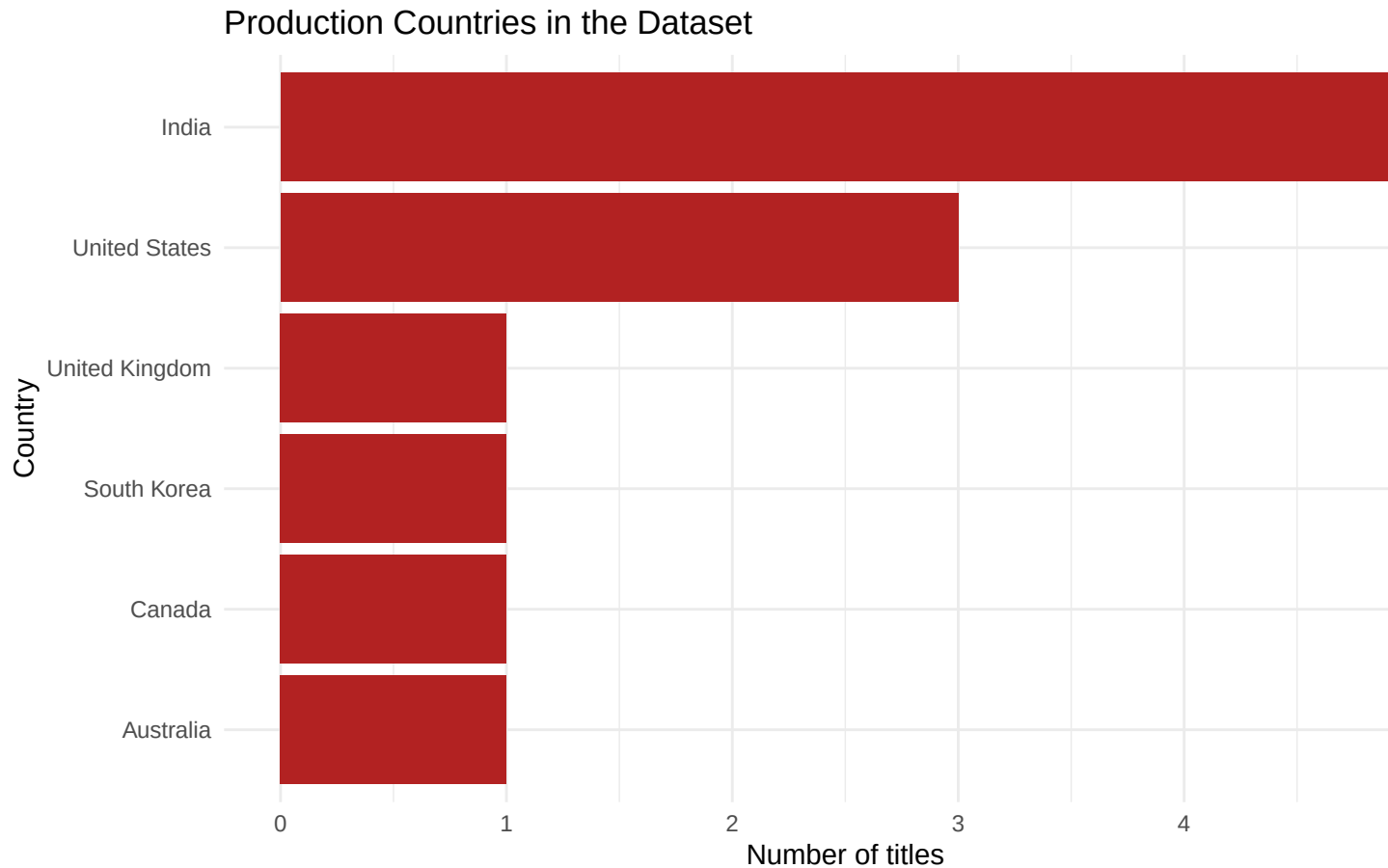
kable(
  top_countries,
  caption = "Production Countries in the Dataset"
)
```

Table 5: Production Countries in the Dataset

country	n
India	5
United States	3
Australia	1
Canada	1

country	n
South Korea	1
United Kingdom	1

```
ggplot(top_countries, aes(x = reorder(country, n), y = n)) +
  geom_col(fill = "firebrick") +
  coord_flip() +
  labs(
    title = "Production Countries in the Dataset",
    x = "Country",
    y = "Number of titles"
  ) +
  theme_minimal()
```



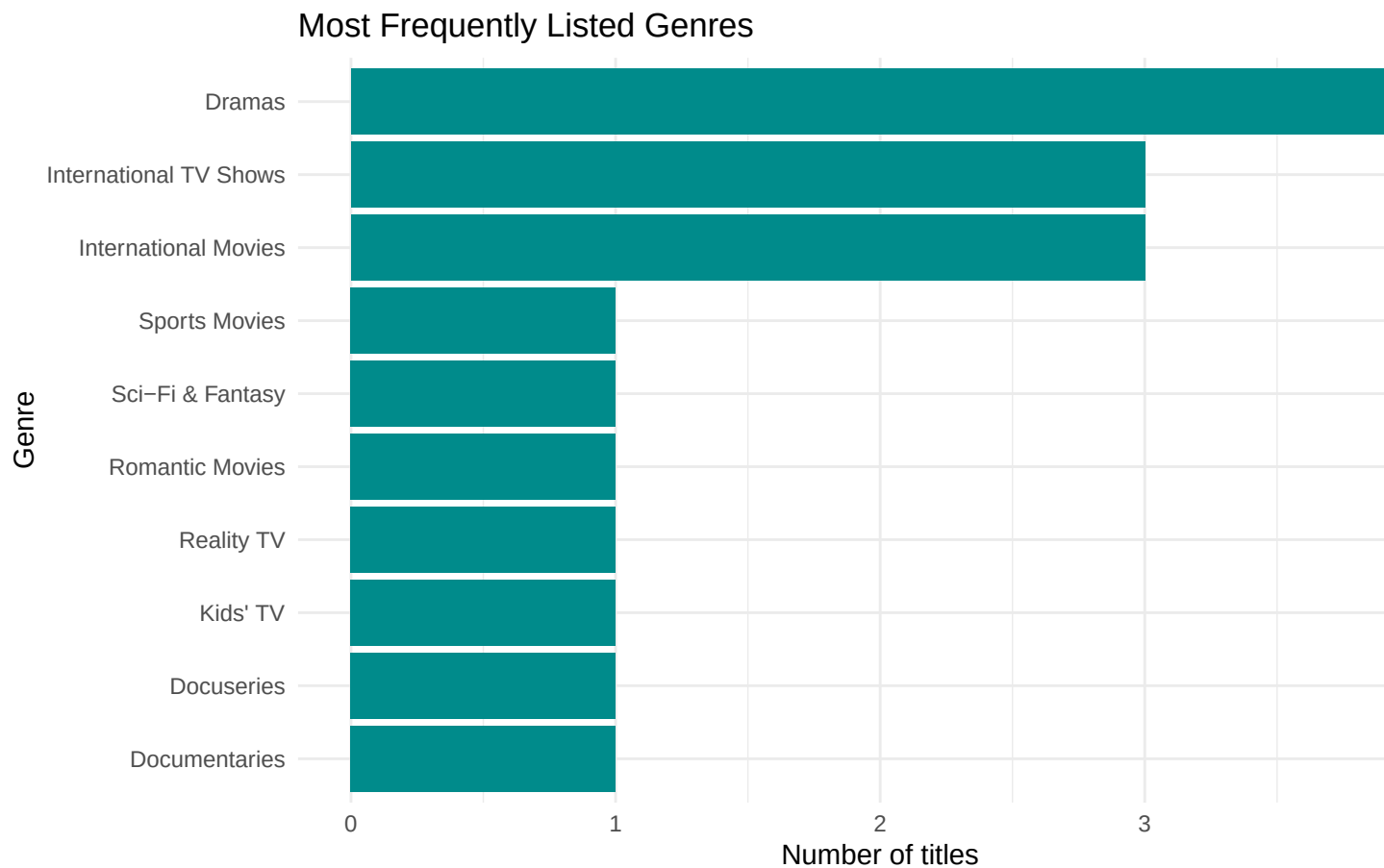
```
top_genres <- netflix_clean %>%
  filter(listed_in != "Unknown") %>%
  mutate(genre = str_split(listed_in, "\\s*")) %>%
  tidyr::unnest(genre) %>%
  count(genre, sort = TRUE) %>%
  slice_head(n = 10)
```

```
kable(top_genres, caption = "Most Frequently Listed Genres")
```

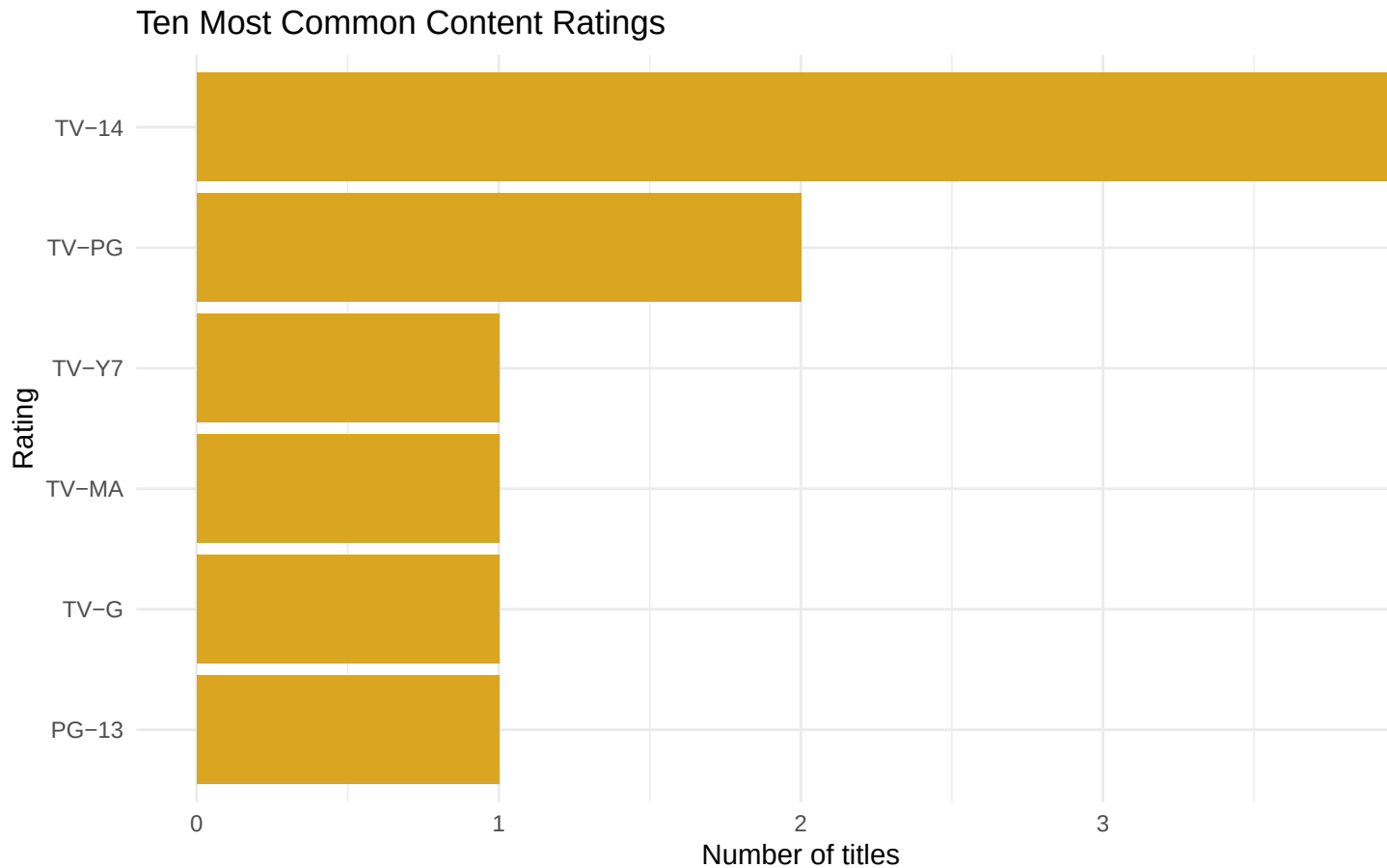
Table 6: Most Frequently Listed Genres

genre	n
Dramas	4
International Movies	3
International TV Shows	3
Documentaries	1
Docuseries	1
Kids' TV	1
Reality TV	1
Romantic Movies	1
Sci-Fi & Fantasy	1
Sports Movies	1

```
ggplot(top_genres, aes(x = reorder(genre, n), y = n)) +
  geom_col(fill = "darkcyan") + coord_flip() +
  labs(title = "Most Frequently Listed Genres", x = "Genre", y = "Number of titles") +
  theme_minimal()
```



```
top_ratings <- netflix_clean %>% count(rating, sort = TRUE) %>% slice_head(n = 10)
ggplot(top_ratings, aes(x = reorder(rating, n), y = n)) +
  geom_col(fill = "goldenrod") + coord_flip() +
  labs(title = "Ten Most Common Content Ratings", x = "Rating", y = "Number of titles") +
  theme_minimal()
```



Key Observations

- The dataset contains 10 titles, including 6 Movies and 4 TV Shows.
- Movies represent the larger content category in the selected dataset.
- The release years represented in the dataset range from 2018 to 2024.
- TV-14 is the most frequently occurring content rating.
- India is the most frequently represented production country after separating multiple-country entries.
- Dramas is the most frequently listed genre after separating multiple-genre entries.

Interpretation of results

- The content-type chart directly answers whether Movies or TV Shows dominate the selected Netflix catalogue.

- Country and genre fields can contain more than one value per title; this report separates comma-delimited values before counting, so a co-produced title can contribute to multiple countries or genres.
- The release-year plot describes titles available in the dataset, not Netflix subscriber demand, viewing hours, or revenue.
- Missing metadata is labelled **Unknown**; therefore, the analysis makes data quality visible instead of treating missing information as a real category.

Findings

- The overview table reports the total number of titles and the split between Movies and TV Shows.
- The release-year chart highlights periods with more titles represented in the catalogue.
- The country and genre charts identify the most frequently represented production locations and categories.
- Results depend on the date, country, and source of the downloaded Netflix dataset; the catalogue changes over time.

Conclusion

This project successfully applies data import, preprocessing, descriptive analysis, and data visualisation to a Netflix catalogue dataset. The analysis provides a useful overview of content type, release periods, ratings, production countries, and genres. A future extension could analyse the relationship between content rating and genre, compare Movies with TV Shows by country, or join the data with IMDb/TMDb ratings where permitted.

Reproducibility checklist

Reproducibility Checklist

1. Place `netflix_titles.csv` and the `.Rmd` file in the same folder.
2. Install the required R packages: `rmarkdown`, `knitr`, `dplyr`, `ggplot2`, `stringr`, and `tidyr`.
3. Open the `.Rmd` file in RStudio.
4. Select **Knit** → **Knit to PDF**.
5. Verify that all tables, graphs, and sections are displayed correctly.